Sham Satish Thakare

Candidate for PhD in Computer Science · Reliable RLVR, Foundation Model Reasoning & Distributed Consensus

Pre-Doctoral Researcher, Capital One Systems Research (Mentored by Yuri Shelke, Distinguished Engineer) · APCOER Pune (SPPU)

Pune, India • shamthakare3000@gmail.com • CS Dossier • GitHub (@shamddd) • SSRN (Author 13130226)

RESEARCH INTERESTS

Reliable Reinforcement Learning (RLVR): Policy optimization (GRPO/PPO), verifiable reward credit assignment, uncertainty calibration, length confounding, and reasoning trajectory geometry.
Fault-Tolerant Distributed Systems: State-machine replication (Raft, Paxos), dynamic quorum adaptation under network partitioning, linearizability verification, and tail latency elimination in modern C++20.
Counterfactual Evaluation: Difference-in-differences state continuation protocols isolating nominal self-correction from unconditional generative capacity.

EDUCATION & HONORS

APCOER Pune (Savitribai Phule Pune University), Pune, India — 2020 – 2024

Bachelor of Engineering (B.E.) in AI & DS | CGPA: 8.70 / 10.00 (First Class with Distinction) — Final Year SGPA: 9.53 / 10.00 (Rank 1)

Honors & Awards: Gold Medal in Mathematics (2024); Department Rank 1; First Class with Distinction (170/170 credits, Seat No: B191102032, PRN: 72159719L, 125th Convocation).

Grade O (Outstanding 10/10) Coursework: Quantum AI (Course 417523A, 10/10), Distributed Computing (Course 417531, 10/10), Artificial Neural Networks (Course 317531, 10/10), Operating Systems (Course 217521, 10/10), Cyber Security (Course 317530, 10/10), Senior Capstone Project (10/10).

Medium of Instruction: Verified on official university transcripts as **ENGLISH** (Ref: EXAM./TRANS./BEN241203044).

Pre-University: Class XII HSC (71.23%, Seat: V001345) • Class X SSC (**85.20% Distinction, Mathematics: 95/100**, Seat: H006786).

RESEARCH & PRE-DOCTORAL EXPERIENCE

Capital One Systems Research, Pune, India — 06/2024 – Present

Pre-Doctoral Researcher | Mentorship: Yuri Shelke (Distinguished Engineer)

Architected and implemented <code>quorumshift</code> in modern C++20: dynamic quorum re-weighting with Delta-bounded clock drift, eliminating p99 tail latency spikes during asymmetric cloud network partitions.

Designed 5-way matched ablation studies on GSM8K/MATH demonstrating that token predictive entropy acts as an unintended length penalty in GRPO (r = +0.486, p < 10^-6).

Engineered asynchronous, lock-free C++20 distributed benchmarking harness testing consensus throughput under chaotic network fault injections.

APCOER Pune (Savitribai Phule Pune University), Pune, India — 06/2023 – 05/2024

Undergraduate Researcher & Peer Academic Mentor

Formulated <code>recovery_eval</code> and <code>stateshift</code> to measure state-contrast continuation in LLM reasoning rollouts.

Conducted recitations and lab tutorials in Data Structures, Algorithms, and Discrete Mathematics for 120+ engineering undergraduates.

PUBLICATIONS & MANUSCRIPTS UNDER REVIEW

• When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in Language Model RL

Sham Satish Thakare. Under Review at IEEE Transactions on Artificial Intelligence (IEEE TAI), 2024. [Code: github.com/shamddd/ear_grpo_reasoning]

- **AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus**

  Sham Satish Thakare, Yuri Shelke (Distinguished Engineer, Capital One). Under Review at IEEE TAI, 2024. [Code (C++20): github.com/shamddd/quorumshift]

- **recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning**

  Sham Satish Thakare. Under Review at IEEE BigData, 2024. [Paper: shamddd.github.io/shamthakare.github.io/pdfs/recovery-eval.pdf]

- **MediRush-SafeAgent: Layered Runtime Enforcement for Healthcare-Commerce LLM Tool Execution**

  Sham Satish Thakare. In Peer Review at Intelligence-Based Medicine (Elsevier) & SSRN Preprint (ID: 7446543), 2024. [SSRN: ssrn.com/abstract=7446543]

TARGET HARVARD SEAS CS FACULTY ALIGNMENTS

Prof. Sham Kakade (Kempner Institute / SEAS): Statistical foundations of reasoning, reinforcement learning policy optimization, and verifiable reward alignment.

Prof. Boaz Barak (SEAS): Theoretical limits and capabilities of deep learning, multi-step reasoning models, and computational complexity.

Prof. Minlan Yu (SEAS): Scalable networked systems, high-speed telemetry, and cloud infrastructure.

Prof. James Mickens (SEAS): Fault-tolerant distributed consensus, state machine replication, and resilient systems.